ENG 2003 -- Effective Engineering Communication

Lassonde School of Engineering, York University

July 19, 2026

ENG 2003 Instructional Team

Lassonde School of Engineering

York University

Re: Hydrogen Fuel-Cell Vehicles: A Clean-Transport Solution for Fleet Applications

Dear ENG 2003 Instructional Team:

I am submitting the attached revised report for Phase 4 (Revise, Finalize, and Reflect) of Term Project 1 titled *Hydrogen Fuel-Cell Vehicles: A Clean-Transport Solution for Fleet Applications*. This topic was approved in July 2026.

The report examines the renewable-energy storage challenge described by Jim Watson, director of the UK Energy Research Centre, and proposes hydrogen fuel-cell electric vehicles (FCEVs) as an engineering solution, primarily in the context of fleet applications such as municipal buses. The report also goes over the progress of the technology, its cost, and its infrastructure preparedness to determine where adoption is most justifiable today. This revised version expands on the Phase 2 draft with additional visual evidence and incorporates feedback received during the peer-assessment phase. A summary of that feedback and the changes made in response appears in Appendix A. I kindly request that you review the report and provide any feedback as soon as possible.

If you have any questions or require further information, please feel free to contact me. Thank you for your time and feedback throughout this project.

Sincerely,

Deven Patel

HYDROGEN FUEL-CELL VEHICLES:

A Clean-Transport Solution for Fleet Applications

Addressing Renewable Energy Storage and Fossil-Fuel Dependence

Deven Patel

ENG 2003 – Effective Engineering Communication

Final Report – Phase 4 (Revise, Finalize and Reflect)

Lassonde School of Engineering – York University

July 19, 2026

Executive Summary

The report examines one of the most significant energy challenges of the 21st century: how to balance electricity supply and demand as more wind and solar power are added to the grid. Jim Watson, director of the UK Energy Research Centre, says this imbalance cannot be fixed without large-scale, long-duration energy storage and that this kind of storage is still in its early stages [1]. Recent numbers illustrate the scale of the problem. In 2024, the California Independent System Operator (CAISO) was forced to reduce wind and solar output by 3.4 million megawatt-hours, a 29% increase over 2023, because the grid could not absorb or store all the power being produced. Texas, Chile, Ireland, and the United Kingdom have also experienced similar cutbacks in renewable output [7], [8] (Figure 1).

To address this problem, the report proposes hydrogen fuel-cell electric vehicles (FCEVs) as an engineering solution, with the strongest focus on commercial fleets such as municipal buses. Extra renewable electricity can be turned into hydrogen through electrolysis, stored for a long time, and later used to power a fuel cell vehicle. This addresses both the storage problem and the transportation sector's reliance on fossil fuels at the same time.

The report finds that FCEV technology is already mature, with vehicles such as the Toyota Mirai and Hyundai Nexo achieving more than 350 miles of range and refuelling in about five minutes, comparable to gasoline cars and faster than battery electric vehicles (BEVs) in terms of refuelling [11]. Fleet operators are already acting on these findings: registrations of hydrogen buses in Europe increased by more than 400% year-on-year in 2025, and large-scale deployments and targets are emerging in China and North America [10]. However, the production of green hydrogen is still estimated to be two to three times more expensive than conventional hydrogen, and public refuelling stations are still limited to a few regions [6], [9].

The report calls for continued investment in hydrogen FCEV technology, with fleet applications being the strongest case at present. The best evidence that the technology works is in fleets such as municipal buses, where rapid refuelling and predictable routes make up for today's cost and infrastructure problems. Passenger-vehicle adoption will follow suit as production costs fall and refuelling infrastructure grows. This recommendation is based on a well-supported technical and deployment case, but it lacks direct proof at the plant level that curtailed renewable electricity is being converted into vehicle hydrogen, an area for future work to close. Closing that gap and advancing this recommendation will require the skills of mechanical engineers in powertrain design, hydrogen storage, and fleet infrastructure.

Table of Contents

List of Figures

List of Tables

1.0 Introduction

Grid operators in California, Texas, and across Europe are now regularly switching off wind and solar farms that could be powering homes, simply because there is nowhere to put the electricity they produce [7], [8]. This situation illustrates the energy storage challenge facing modern electricity grids, identified by Jim Watson, director of the UK Energy Research Centre, as one of the defining energy challenges of the 21st century [1]. As more wind and solar power is added to national grids, operators face more periods where renewable output is higher than demand and periods where it falls short. Without a way to store this surplus energy for long periods, this mismatch between supply and demand cannot be solved [1]. Hydrogen fuel-cell electric vehicles (FCEVs) are proposed in this report as a technology that can absorb this surplus renewable electricity, formally defined as "curtailment" (output reduction) in Section 2.1, by converting it into hydrogen fuel while also reducing the use of fossil fuels in transportation, from cars to commercial fleets like buses and delivery trucks. From a mechanical engineering perspective, the proposal focuses on automotive powertrain design, fuel-cell vehicle architecture, and fleet deployment.

The purpose of this report is to determine whether hydrogen fuel-cell vehicles are a realistic way to help solve the renewable-energy storage problem while also reducing transportation emissions. This is written for a general engineering audience who understand basic energy and transportation concepts but may not know much about hydrogen fuel cells. This report is not a detailed financial feasibility study, and it does not model a specific transit agency's fleet-conversion plan. Instead, it pulls together current engineering and market evidence to build a general, evidence-based case.

The rest of the report is organized as follows. Section 2.0 provides the background needed to understand renewable-energy curtailment and how hydrogen fuel-cell technology works. Section 3.0 looks at the current state of FCEV technology, the vehicles available today, hydrogen production and storage, and fleet-scale deployment. Section 4.0 discusses the strengths and weaknesses of the evidence and recommends where FCEV deployment makes the most sense right now. Section 5.0 concludes the report.

2.0 Background

2.1 The Renewable Energy Storage Challenge

Electricity grids are usually balanced by adjusting the output of dispatchable generators such as natural gas, coal, and hydro plants to match demand in real time. But wind and solar cannot do this balancing as easily, since their output depends on the weather rather than on demand. As grid operators around the world have added more renewable capacity, they've begun to encounter curtailment, or the purposeful reduction of wind or solar generation when the grid can't use or store it. In 2024, the California Independent System Operator (CAISO) curtailed 3.4 million megawatt-hours of wind and solar generation, a 29% increase over 2023, with solar comprising 93% of the curtailed energy [7]. Also in 2024, Texas's ERCOT grid curtailed more than 8 TWh of wind and solar output, while Chile, Ireland, and the United Kingdom experienced similar curtailment rates of 5-15% [7], [8]. The International Energy Agency's Renewables 2024 report says this trend is expected to continue, since renewable capacity is being added faster than grid flexibility and storage [8]. This situation is the same imbalance that Watson describes generation now often peaks in summer because of solar power, rather than in winter when demand is highest, and without large-scale, long-duration storage, this mismatch cannot be solved [1]. Figure 1 summarizes the scale of this problem across the five regions discussed above.

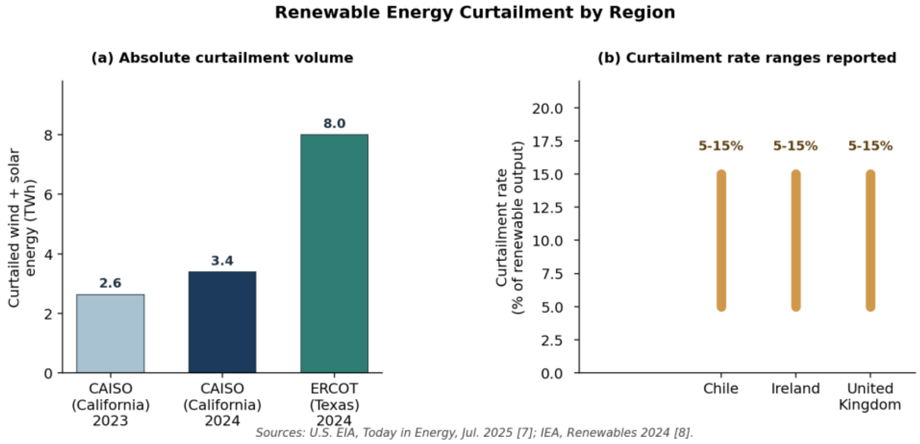

Figure 1: Renewable energy curtailment by region, (a) – Absolute Curtailment Volume, (b) Curtailment Rate Ranges Reported. (data from [7], [8])

Batteries work well for short-duration storage, meaning a few hours, but are limited by cost and energy density for long-duration, seasonal storage [4], the kind of storage Watson argues is needed [1]. Hydrogen offers another way to store energy: surplus renewable electricity can be used to split water into hydrogen and oxygen through electrolysis [4]. The hydrogen gas can be compressed, stored for long periods, and converted back into electricity, heat, or mechanical work as required [4]. Hydrogen obtained in this way is usually called "green hydrogen" if the electricity used is fully renewable [4]. This contrasts with "grey" or "blue" hydrogen, which are made from natural gas without or with carbon capture [4].

2.2 Hydrogen Fuel-Cell Vehicle Basics

This section explains how stored hydrogen is used, with transportation being one of its most developed applications. A hydrogen fuel-cell electric vehicle (FCEV) stores compressed hydrogen gas onboard and runs it through a proton-exchange membrane (PEM) fuel cell. This combines hydrogen with oxygen from the air to generate electricity through a chemical reaction, with water vapour as the only tailpipe emission [2]. Figure 2 illustrates this process schematically.

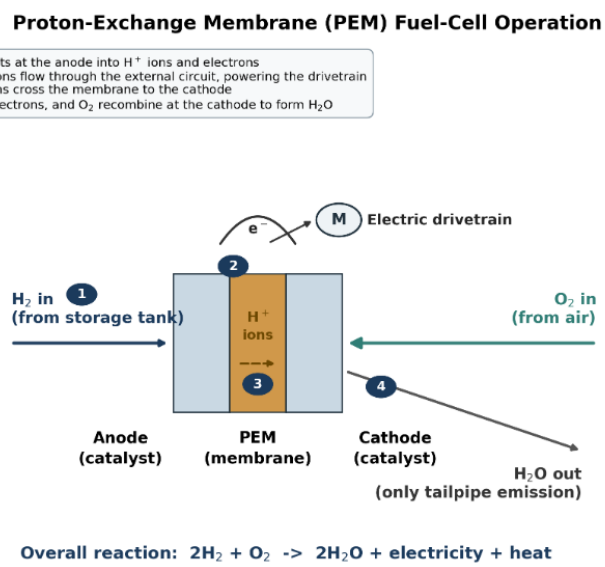

Figure 2: Proton-exchange membrane (PEM) fuel-cell operation (Based on [2]).

This electricity powers an electric drivetrain like a battery electric vehicle (BEV). The main difference is that an FCEV stores its energy as compressed hydrogen gas, which refuels in minutes, instead of storing it in a battery pack, which takes tens of minutes to hours to recharge [2], [3]. Compared to regular gasoline (ICE) vehicles, FCEVs produce no tailpipe emissions of carbon dioxide, nitrogen oxides, or particulate matter. Compared to BEVs, they refuel faster and, for heavier vehicles, offer better range for their weight, since compressed hydrogen tanks are lighter than the large battery packs needed for equal range [2], [3].

Figure 3 summarizes this pathway. FCEVs are examined in this report as a technology that could address Watson's storage challenge and reduce fossil-fuel use in road transportation because they store curtailed renewable electricity as hydrogen and use that hydrogen to power vehicles with zero tailpipe emissions.

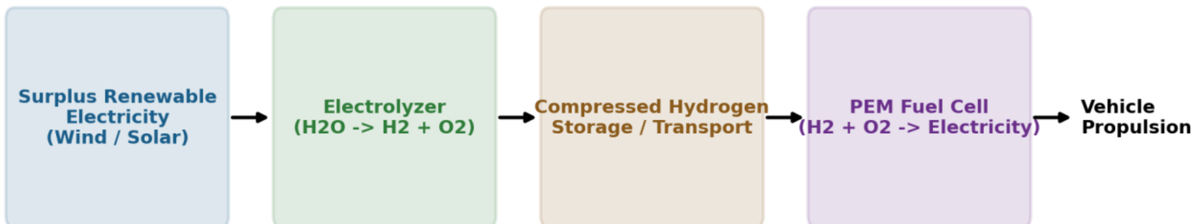

Figure 3: Renewable-to-hydrogen-to-vehicle energy pathway (Based on [4]).

3.0 Main Topics

3.1 Fuel-Cell Vehicle Technology and Commercial Landscape

Consumers have had the option to purchase fuel-cell passenger vehicles for over a decade, though adoption remains limited compared to battery electric vehicles. As of 2026, the Toyota Mirai and Hyundai Nexo are the only two FCEVs still in production and sold in North America. The Honda Clarity Fuel Cell, an earlier model, was discontinued in 2020, and Honda is now developing a plug-in hybrid hydrogen CR-V e:FCEV instead [11]. Table 1 compares the specs of these three vehicles.

Table 1: Comparison of commercial fuel-cell passenger vehicles [11].

Vehicle	Range	Refuel Time	Fuel Economy (MPGe)	Starting Price (USD)
Toyota Mirai (2024)	~402 mi (647 km)	~5 min	67 city / 64 hwy	$50,190
Hyundai Nexo (2024)	354-380 mi (570-611 km)	~5 min	61 city / 59 hwy	$58,785
Honda Clarity Fuel Cell (2021, discontinued)	360 mi (579 km)	~5 min	68 city / 67 hwy	$49,500 (at launch)

All three vehicles refuel in about five minutes, which is like a regular gasoline car and much faster than the 20 minutes (DC fast charging) to several hours (Level 2 charging) usually needed for a BEV with similar range [11]. This refuelling speed is the main practical advantage FCEVs have over BEVs, and it matters more as vehicle use increases, which is discussed further in Section 3.3.

3.2 Hydrogen Production, Storage, and the Cost Barrier

The production method of hydrogen determines the environmental benefit of an FCEV. As introduced in Section 2.1, Gunathilake et al. [4] classify hydrogen production by colour based on feedstock and carbon handling; among these pathways, only green hydrogen actually provides the storage benefit that Watson describes, since it is the only one that uses surplus renewable electricity instead of fossil fuels [1], [4].

Cost is the primary barrier stopping green hydrogen from replacing grey hydrogen at scale. Green hydrogen currently costs an estimated \$3.50-\$6.00 per kilogram to produce (a cited estimate, not modelled independently here), compared to \$1-\$3 per kilogram for grey hydrogen. That gap is shrinking electrolysis equipment costs have dropped roughly 60% since 2010, with another 50% drop projected by 2030 as the technology improves [9]. Onboard storage is also a real limitation. Gunathilake et al. [4] compare four physical storage methods: high-pressure gas, cryogenic liquid, solid-state, and liquid organic hydrogen carriers. They conclude that compressed high-pressure gas at 700 bars, the method used in the Mirai and Nexo, currently provides the best balance of weight, cost, and refuelling speed, despite needing reinforced tanks and dedicated fueling infrastructure [4], [6].

Refuelling infrastructure is another constraint that is still developing. According to the U.S. Department of Energy's Alternative Fuels Data Centre, hydrogen fueling stations are still mostly concentrated in California, with new stations only starting to appear in Hawaii and the northeastern United States. This growth is supported by incentives like the federal 45V production tax credit and updated safety codes (NFPA 2) [6]. Because of this, FCEVs are currently only practical in or near these limited refuelling areas.

3.3 Fleet-Scale Deployment: Buses and Depot-Based Fleets

Passenger-vehicle adoption of FCEVs has been slow, but fleet applications, especially public transit buses, are growing fast, showing where hydrogen's advantages matter most. Wijayasekera et al. [5] argue municipal bus fleets are an especially suitable fit for hydrogen: fixed routes make fuel demand predictable, a single central depot can refuel the whole fleet, and heavy daily use makes rapid refuelling more valuable than the multi-hour recharge times of large bus batteries. These are precisely the conditions where FCEVs' main advantage, rapid refuelling, outweighs their main disadvantage: the cost and complexity of producing and distributing hydrogen [3], [5].

Deployment data collected by the industry publication Sustainable Bus illustrates this trend [10]. In Europe, 279 hydrogen buses were registered in the first half of 2025 alone (+426% year-over-year); Solaris alone delivered 225 buses in 2024, including a 130-bus order for Bologna, Italy. In China, Nanning announced plans to convert its entire 7,000-bus battery-electric fleet to hydrogen fuel-cell/battery hybrids, citing performance advantages at scale. North American deployment is smaller but growing: Ballard announced a 200-engine (~20 MW) order to New Flyer in November 2024, and New York's MTA plans to put its first two hydrogen buses into service in summer 2026, aiming for up to 35 eventually [10]. Table 2 summarizes this regional deployment pattern, and Figure 4 visualizes the same figures on a log scale to show the difference in scale between an announced target and current deployment activity.

Table 2: Snapshot of hydrogen fuel-cell bus deployment activity and its primary driver, by region, 2024-2026 [10].

Region	Recent Deployment Activity	Notable Driver
Europe	279 buses registered H1 2025 (+426% YoY); Solaris delivered 225 buses in 2024	Manufacturer-led tenders (e.g., 130-bus Bologna order)
China	Nanning targeting full conversion of 7,000-bus BEV fleet to FCEV/hybrid	National fuel-cell vehicle targets
North America	Ballard/New Flyer 200-engine order (2024); MTA piloting first 2 buses (2026)	State/federal clean-transit incentives

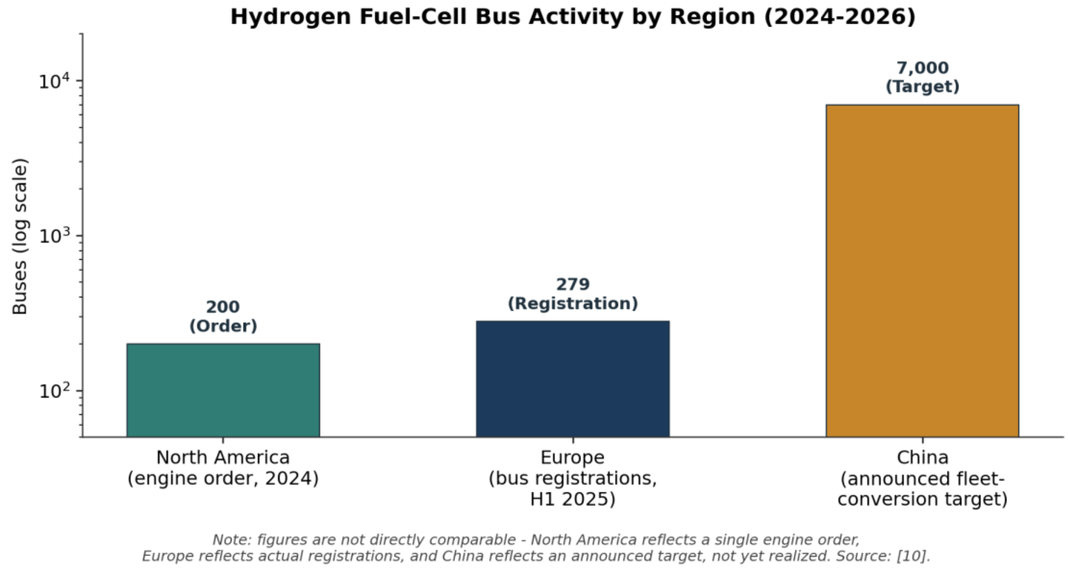

Figure 4: Hydrogen fuel-cell bus activity by region, 2024-2026 (data from [10]).

Trucks and other depot-based commercial fleets, such as delivery vans, refuse trucks, and regional freight, share similar traits with buses: fixed routes, centralized refuelling, and heavy daily use. These are often described in the literature as the next logical step for fleet-based FCEV deployment once bus programs prove the concept works [3], [5].

4.0 Discussion

The evidence gathered in Section 3.0 supports Watson's claim that renewable energy storage is a real and growing problem and supports the idea that hydrogen and FCEVs can help with it, but only under certain conditions. Watson's framing of the problem is more convincing as curtailment data from CAISO, ERCOT, and the IEA (Figure 1) show the storage problem he describes is not a UK-specific problem but a pattern emerging across multiple grids with different regulations [1], [7], [8]. However, none of the FCEV-specific sources used in this report measure how much curtailed renewable electricity is being turned into hydrogen for transportation, as opposed to other uses like industrial feedstock or blending into the natural gas grid. This is a real gap in the evidence: the idea that curtailment exists and therefore hydrogen vehicles are absorbing it makes logical sense but has not been proven at scale. A stronger version of this report would trace curtailed grid electricity all the way through to vehicle fuelling using plant-level data.

The vehicle-level and fleet-level evidence is more direct. The comparison in Table 1 shows that FCEVs already match or beat the range of most BEVs while refuelling about as fast as gasoline vehicles, which is a real engineering advantage happening now, not just a projected one [11]. The fleet deployment data in Table 2 and Figure 4 show operators are already acting on this advantage: hydrogen bus registrations grew over 400% year-over-year in Europe, and operators in China and North America are making real purchasing decisions and announcing targets based on the same fixed-route, centralized-refuelling logic that Wijayasekera et al. [5] describe [5], [10]. On the other hand, the cost data in Section 3.2 is the clearest weakness in the case for FCEVs: green hydrogen still costs two to three times more to produce than the grey hydrogen that makes up most of today's supply [9], and public refuelling stations outside dedicated fleet depots are still sparse and limited to a few regions [6]. A passenger-vehicle buyer choosing between a Mirai or Nexo and a BEV faces real infrastructure risk that a bus operator refuelling from a single depot station does not.

Altogether, this evidence supports hydrogen FCEVs as a technology that works well for clean transport, especially in fleet settings, rather than as a universal replacement for all vehicles. The vehicle-level data in Table 1 show that the technology already performs reliably for passenger use, matching or beating BEV range while refuelling in minutes [11]. The fleet deployment data in Table 2 and Figure 4 show that depot-based fleets, such as municipal buses and eventually delivery trucks and regional freight as the technology matures, are the fastest-moving proof of this. This is because they avoid FCEVs' weakest point, public refuelling infrastructure, while taking full advantage of their strongest point, rapid refuelling under heavy daily use [5]. Passenger-vehicle adoption is likely to follow the same path, just over a longer timeline tied to the cost and infrastructure trends discussed in Section 3.2.

One limitation of this report should be pointed out: it does not do an independent techno-economic or life-cycle emissions calculation and instead relies on figures pulled together from the cited sources. A more detailed version of this analysis, suitable for a final feasibility study, would model the total cost of ownership for a specific fleet size and route and calculate life-cycle emissions for green, blue, and grey hydrogen specifically for vehicle fuel use.

5.0 Conclusion

This report looked at the renewable-energy storage challenge identified by Jim Watson of the UK Energy Research Centre and proposed hydrogen fuel-cell electric vehicles, for both passenger and commercial-fleet use, as a mechanical engineering technology that could address it. Renewable curtailment data from California, Texas, and the IEA confirms that the storage problem Watson describes is real and getting worse as renewable capacity grows [1], [7], [8]. Fuel-cell technology, shown by commercially available vehicles like the Toyota Mirai and Hyundai Nexo, is already mature and offers refuelling speeds that battery electric vehicles cannot match [11]. Fleet operators all over the world are already using these results, and the number of hydrogen buses on the road is growing quickly in Europe, China, and North America [10]. At the same time, the cost of producing truly low-carbon green hydrogen and the limited amount of public refuelling infrastructure are real problems that are not yet solved [6], [9].

Overall, the evidence most strongly supports continued investment in hydrogen FCEV technology for fleet applications specifically, where fixed routes and centralized refuelling already make up for the cost and infrastructure problems described in Section 3.2. Municipal buses provide the clearest near-term proof that the technology works under real conditions, and trucks and other depot-based fleets are a logical next step. Passenger-vehicle adoption is expected to follow the same path over a longer timeline, depending on falling green-hydrogen costs and growing refuelling infrastructure. This trend makes hydrogen FCEVs a technology that complements battery electric vehicles, with mechanical engineers in powertrain design, hydrogen storage, and fleet infrastructure having an immediate role to play in the fleet segment. As noted in Section 4.0, this conclusion is based on a well-supported technical and deployment case, not a direct measurement of curtailed electricity flowing into vehicle hydrogen at the plant level. This is a gap that future work should close.

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Appendix A: Summary of Peer Feedback and Revisions

This report received feedback from three peers during Phase 3 (Assess) via PeerScholar. All three reviewers rated Logic of the Proposal, Use of Evidence and Reasoning, and Structure as "Great"; two of three rated Visual Elements as "Great" and one as "Satisfactory"; and all three rated Mechanics as "Satisfactory."

Reviewer 1 praised the report's precise, fleet-focused recommendation and balanced discussion of limitations but observed proofreading issues and awkward phrasing in the letter of transmittal and introduction and requested better transitions between parts. The letter of transmittal was rewritten to prevent a redundant repetition of the full report title; the opening paragraph of the introduction was rewritten for clarity, e.g., the awkward construction "This situation is the energy storage challenge" was replaced with "This situation illustrates the energy storage challenge"; and an explicit bridging sentence was added at the beginning of Section 2.2 to connect it back to Section 2.1.

Reviewer 2 identified a grammar error on page 1 ("reducing us of fossil fuel in transportation," corrected to "reducing the use of fossil fuel in transportation") and suggested shortening the title for readability. Both were addressed: the grammar error was corrected, and the title was shortened from "Hydrogen Fuel-Cell Vehicles as a Targeted Clean-Transport Solution: Addressing Renewable Energy Storage and Fossil-Fuel Dependence in Fleet Applications" to "Hydrogen Fuel-Cell Vehicles: A Clean-Transport Solution for Fleet Applications," with the storage/fossil-fuel framing kept as a subtitle. Reviewer 2 also noted that additional visual elements would strengthen the report; this revision adds three new figures (the curtailment chart, the PEM fuel-cell diagram, and the bus-deployment chart) beyond the single figure in the Phase 2 draft.

Reviewer 3 had three points. Firstly, the term "curtailment" was introduced in the introduction and then formally defined in Section 2.1. The reviewer found this confusing, which led to changing the introduction to use general terminology ("surplus renewable electricity") and to forward-reference Section 2.1 for the formal definition. Secondly, the colour-based classification of hydrogen (green/grey/blue) was completely highlighted twice, once in Section 2.1 and once in Section 3.2. The Section 3.2 explanation was reduced to pointing to Section 2.1 instead of reiterating it. Lastly, the reviewer felt that the phrase "rather than a peer-reviewed study" to describe sources took away from the credibility of the report itself, while the report was arguing that its evidence was strong. This phrasing was removed from the note of Table 1 and from Section 3.3 and was replaced with neutral framing.

None of the three reviewers contradict each other, but instead each addressed a different area of the report (mechanics and transitions, title and grammar, and terminology and source framing, respectively), so no input was required to be reconciled or counterargued. Beyond the individual PeerScholar comments, this version additionally improved the report overall by including three new figures and a separate List of Figures and List of Tables, as per the Phase 4 directive to go beyond the peer review comments alone.

An additional data appendix was considered but rejected, since the only candidate table merely repeated values already presented in the text of Section 2.1 and in Figure 1. Keeping that information directly in the body of the report, rather than pushing it into a separate appendix the reader would need to flip to, kept the evidence integrated with the argument it supported. In addition, since the text and Figure 1 already substantiate the claim on their own, a standalone data table would have added redundancy rather than additional support.

Every section of the report was reviewed and altered during the revision. Some received structural and content changes, while the remaining sections, including the References list and Table 1's vehicle specifications, received lighter editing for grammar, wording, and consistency with the rest of the report. No section remains in its original Phase 2 form.

AI Use Disclosure Statement

Artificial intelligence helps refine my English and correct grammar errors as academic writing at this level may be hard to get exactly perfect. It also helped me express my ideas more clearly, not just by correcting mistakes, but by suggesting better ways to say what I was trying to say, especially where my original phrases were too vague or badly constructed. Seeing the original phrases next to the AI's proposed versions helped me realize where I was going wrong in my own work, and to acquire better sentence structure and word choice that I may carry forward to future tasks, rather than merely accepting a corrected version without learning from it. The AI was also useful in other ways, not just for language. When I had a rough idea of what I wanted to argue but struggled to structure it clearly, I used it to talk through the logic of my argument and check whether my reasoning flowed in a sensible order. This helped me to catch gaps in my own thinking early, before they became bigger problems in the final draft. I also used it to check formatting and citation alignment with the needed style guide, which saved me time to spend into making the actual content of my analysis stronger. All ideas, arguments, analysis and conclusions in this report are my own. AI was simply used as a tool to assist refine the language, organize my thoughts and keep me on track.